

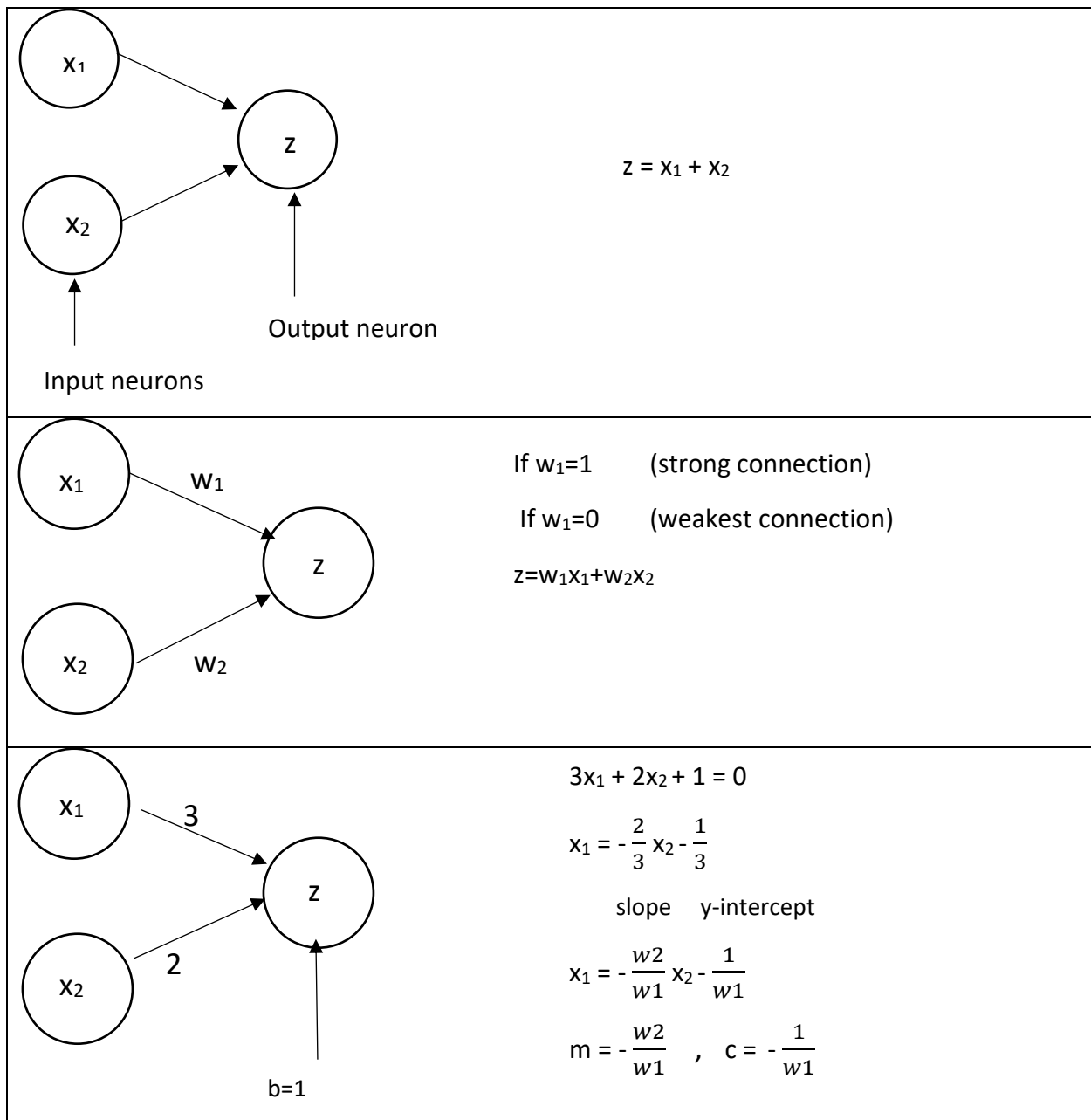
ANN and DL (CAI-361)

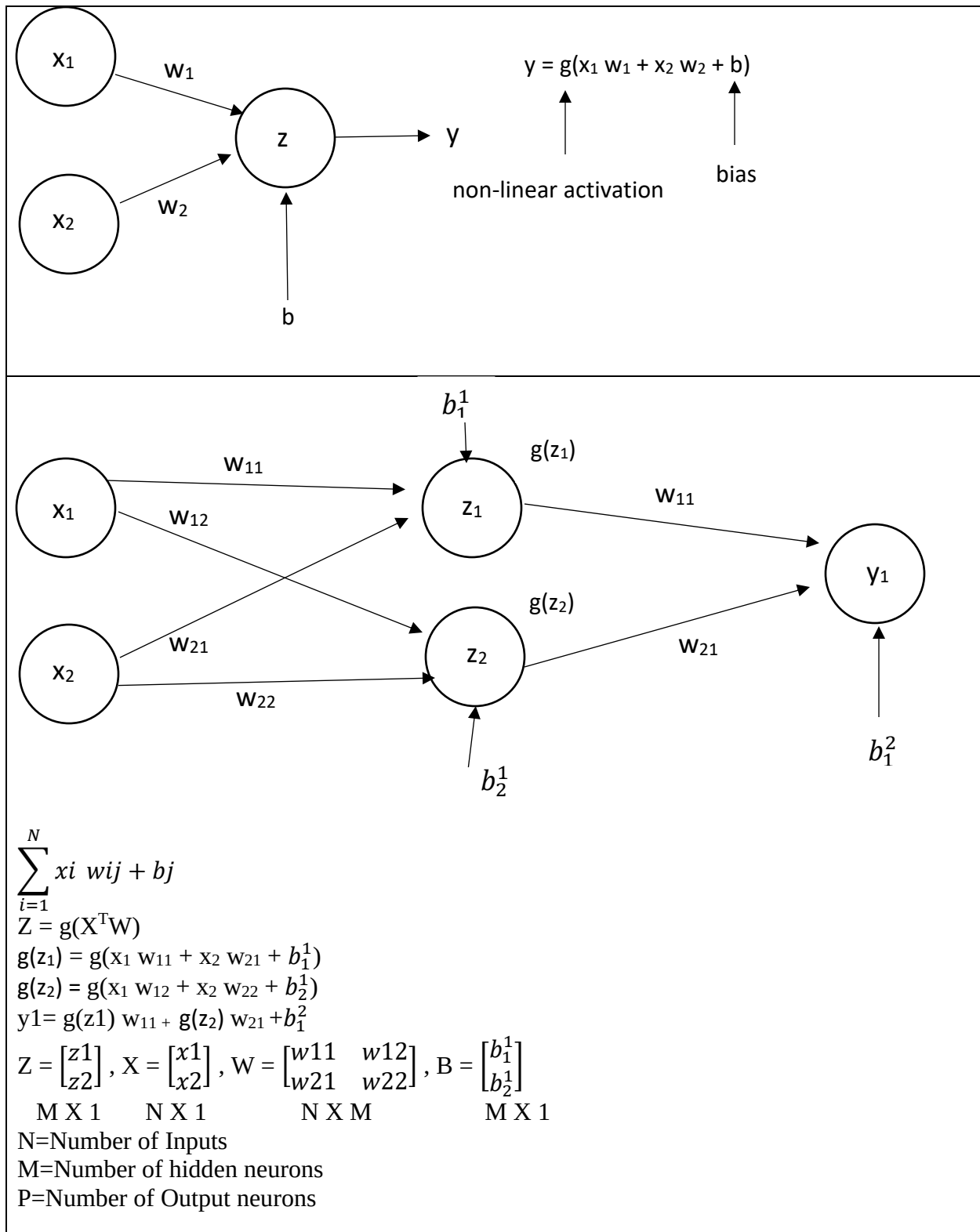
Objectives:

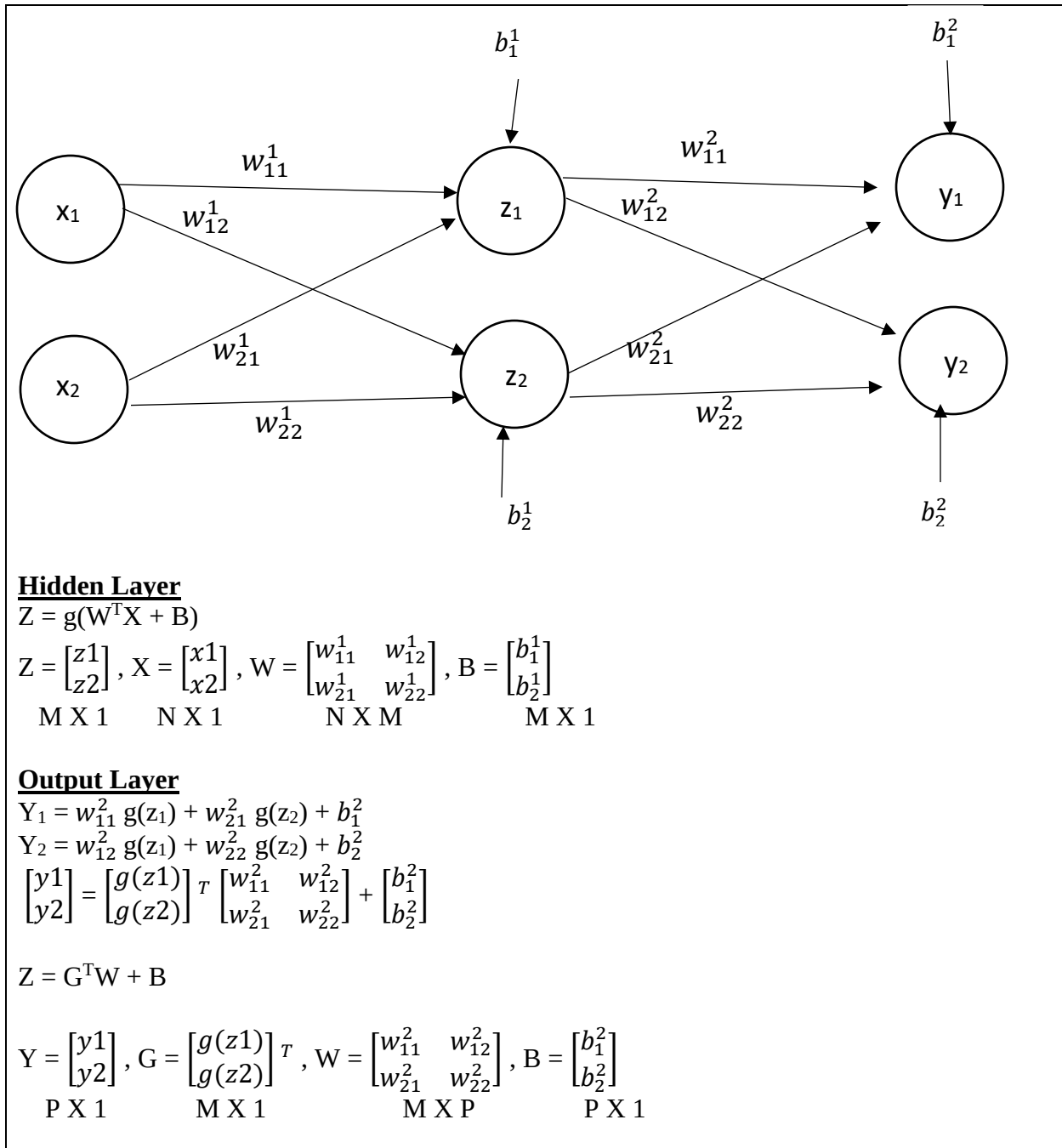
- Artificial Neural Networks
- Feed Forward Neural Networks
- Activation Functions

Artificial Neural Networks:

Neural Networks are based on loosely how brain works. Building block of neuron is given below explained from simple to complex structures.







Feed Forward Neural Network:

Perceptron is a single layer neural network and a multi-layer perceptron is called Neural Networks. The feedforward neural network was the first and simplest type of artificial neural network devised. In this network, the information moves in only one direction—forward—from the input nodes, through the hidden nodes (if any) and to the output nodes. There are no cycles or loops in the network. The basic components of a neural network are:

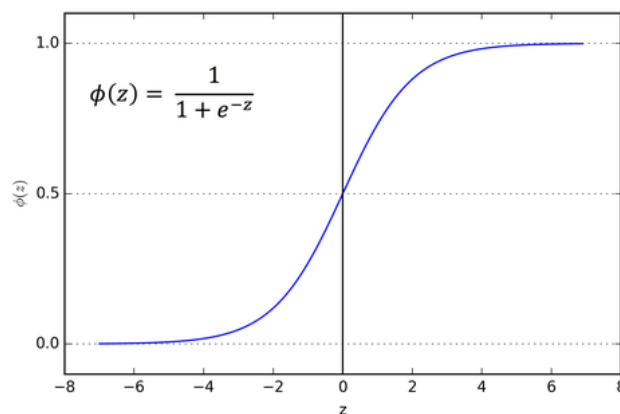
1. **Input Layer:** The input layer consists of one or more input neurons, which receive input signals from the external world.
1. **Weights:** Each input neuron is associated with a weight, which represents the strength of the connection between the input neuron and the output neuron.
2. **Bias:** A bias term is added to the input layer to provide the network with additional flexibility in modelling complex patterns in the input data.
3. **Activation Function:** The activation function determines the output of the perceptron based on the weighted sum of the inputs and the bias term. Common activation functions used are step function, sigmoid function, and ReLU function.
4. **Output:** The output of the perceptron is a single binary value, either 0 or 1, which indicates the class or category to which the input data belongs.
5. **Training Algorithm:** The neural network is typically trained using a supervised learning algorithm such as the perceptron learning algorithm or backpropagation. During training, the weights and biases of the network are adjusted to minimize the error between the predicted output and the true output for a given set of training examples.

Activation Functions:

It is used to determine the output of neural network like yes or no. It maps the resulting values in between 0 to 1 or -1 to 1 etc. (depending upon the function).

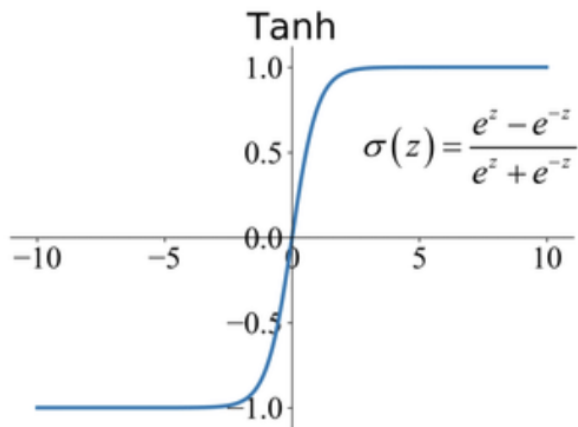
1. Sigmoid:

The main reason why we use sigmoid function is because it exists between (0 to 1). Therefore, it is especially used for models where we have to predict the probability as an output.



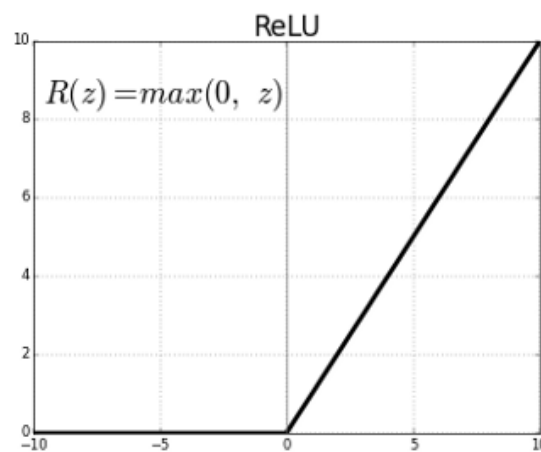
2. hyperbolic tangent (tanh) Activation Function

The tangent hyperbolic function simply referred to as the **tanh** function. The output range of the tanh function is (-1, 1) and presents a similar behavior with the sigmoid function.



3. ReLU (Rectified Linear Unit) Activation Function

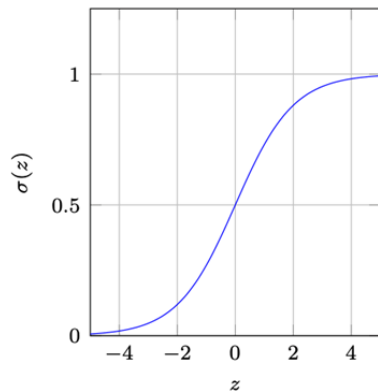
The ReLU is the most used activation function in the world right now. It is used in almost all the convolutional neural networks or deep learning. As you can see, the ReLU is half rectified (from bottom). $f(z)$ is zero when z is less than zero and $f(z)$ is equal to z when z is above or equal to zero.



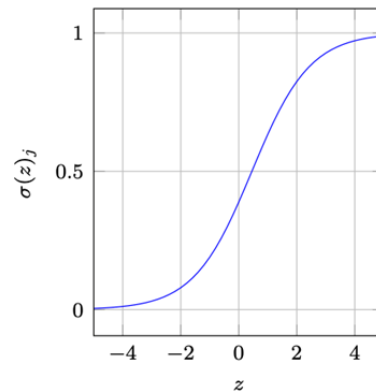
4. Softmax

The softmax activation function transforms the raw outputs of the neural network into a vector of probabilities, essentially a probability distribution over the input classes. Consider a multiclass classification problem with N classes. The softmax activation returns an output vector that is N entries long, with the entry at index i corresponding to the probability of a particular input belonging to the class i . Here K is the number of classes in multi class classification.

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$



(a) Sigmoid activation function.



(b) Softmax activation function.

Figure 1: Sigmoid and Softmax activation functions

Characteristics of Activation Functions

- Sigmoid is used for Binary Classification.
- The probabilities sum of sigmoid does not need to be 1
- Softmax is used for Multi-classification.
- The probabilities of softmax sum will be 1.
- Sigmoid can lead us to gradient decent problem where the updates are so low.
- It turns out that the tanh activation usually works better than sigmoid activation function for hidden units because the mean of its output is closer to zero, and so it centers the data better for the next layer.
- Sigmoid or Tanh function disadvantage is that if the input is too small or too high, the slope will be near zero which will cause us the gradient decent problem.
- If we remove the activation function from our network that can be called linear activation function. (Can classify only linearly separable data)
- Activation functions add non-linearity in the data.
- We can remove activation function in the output layer if the output is real numbers (regression problem). But even in this case if the output value is non-negative we could use RELU instead.

Practice Task1:

Consider a sentimental analysis system on the text. The system can take the Twitter text as input and tells if the text has happy, sad, angry, or neutral emotions. Each Twitter text is preprocessed and encoded into a vector of fixed length of 500. Consider a neural network with three hidden layers each containing 50, 100, and 150 neurons. **Find total learnable parameters there. Write three hidden layers and one output layer equation**, clearly indicating size of each vector and matrix in the equation.

Practice Task2:

Consider the dataset of people who are doing their jobs in Islamabad, the dataset has columns for qualification, years of experience, the salary they get, the job sector in which they are doing the job, and

their designations. The classifier provides output appropriate, excellent, and below average to each salary drawn by a certain employee. (Use 5 Neurons in the hidden layer).

(a) Write the close form of the Hidden layer and output equations, and also show the sizes of vectors and matrices in each case.

(b) Do you think that if there are some features that do not play any role in the classification and classification could be done the same even if they are removed? Considering 5 Neurons in the hidden layer, how many weights do you think the network needs to train overall?

(c) The class and predicted columns have a string to represent each class while to write the loss function we need to use a number what strategy do you think is appropriate for class and predicted columns for the loss function? Also, write the loss function for the table and give a numerical value for the loss function.

Qualification	Experience	Job Sector	Designation	Salary	Classes	Predicted
Graduation	5	Petroleum	Site Engineer	5,30,000	excellent	excellent
Post Graduation	2	Telecommunication	Site Support Engineer	85,000	excellent	appropriate
Intermediate	1	Telecommunication	Clerk	20,000	below average	below average
Matriculation	6	Education	Guard	25,000	appropriate	appropriate
Intermediate	15	Banking	Casher	65,000	excellent	appropriate
Post Graduation	12	Education	Associate Professor	3,50,000	excellent	excellent
Graduation	10	Telecommunication	Manager	2,50,000	appropriate	below average
Graduation	12	Education	Assistant Professor	2,10,000	below average	appropriate

Practice Task3:

Assume you are training a simple neural network for binary classification, where both hidden layers make use of an activation function. What is the output mean squared error of one forward pass using the neural network and initializations below. Round your result to a number **with 4** decimals. (Answer: 0.4896)

